

munication, high frame-rate video capture and display, greater than 4K resolution videos and a number of processing cores are on everyone's mind. High-definition video resolutions start from 1080p and go up to 3Gbps of high-definition serial digital interface."

He suggests that there are various techniques available to improve audio-visual quality. Once we convert all audio-visual signals from analogue to digital form, quality will improve and noise on the signal can be easily removed. "For audio signals, in particular, more types of

explains Giovino.

Explaining the concept of memory, Bharat says, "There are two types of memories used in various processor based solutions. The current trend is permanent memory like embedded-multimedia-card, serial advanced technology attachment and solid-state memory, which are small in physical size but offer large storage capacities. Multi-chip package and package on package memories are also popular in mobile and consumer electronics applications."

Adding to this, Giovino goes into more detail, "Larger amounts

would be to design transistors of such small dimensions. "More transistors within a given area lead to more capabilities, and it is a matter of choosing the right design process," he adds.

Newer materials used in manufacturing are also influencing changes for more suitable semiconductor designs, feels Srinath.

Enhancing sensing

A media processor processes the inputs it gets, and hence sensors become vital. A few qualities expected in such sensors, from Bharat's experience, are support for night vision, optical/digital zoom, area of interest, ambient light sensing, auto-detection of objects, filter option for video based noise and real-time data collection.

"Built-in analogue-to-digital converters, digital signal processors, digital interfaces instead of analogue interfaces, and outputs are as important," chips in Srinath.

Software is king

Most of the latest processing techniques are possible thanks to enhancements in software content and programs. "One of the most popular video-processing techniques, namely, video analytics, is implemented via software," says Bharat. "Most media-processing engines are supported with video accelerators, encode and decode engines," he adds.

Code, and decode the code. Incremental innovation is what drives codecs, says Chandak.

Bharat adds, "Video encode and decode are implemented using H.264, Moving Picture Experts Group 2 and 4 standards. Audio encode and decode are implemented using MP3, Windows Media Audio, Advanced Audio Coding standard techniques. H.265 and lossy video compression are some of the new optimisation techniques we have seen in the past year."

From screen to touchscreen, and now gestures

Manuel Rei, industry director, Dassault Systemes, speaks about how trends have evolved over the years. "Ten years ago, the means to communicate with your system was by using a keyboard or a mouse. Slowly, the scenario changed to using your hands to directly access files, with the introduction of touchscreen; phones making the most of this feature. The world has now become fancier; processors now need to recognise an eye contact or movement of a finger. Soon, we will see more gesture based devices." Are we going to have intuition based processing next?

Pick and choose the processor for your application

"A buyer has to be very clear with his target application, its power budget, commercial budget, feature set requirement and footprint/area requirements and so on. After all these factors are fulfilled, performance and functionality of the processor gains a lot of importance. Any target processor needs to meet the functional and performance requirements to be suitable for the target application," says Nate Srinath, founder-director, Inxee Systems Pvt Ltd.

digital filtering and sound synthesis is being implemented on-chip," according to Giovino.

Real-time processing. "Time-critical events call for efficiently managing time synchronisation, latency and throughput," says Bharat.

"To overcome this issue, we need high-speed processing and communication infrastructure. We also need to have very good audio-visual file-compression algorithms to reduce file size," he adds.

Their storage space, the memory. "The desired features in a media processor are fast cores that can handle math-intensive graphical processing, along with plenty of random access memory (RAM),"

of on-chip memory, or fast external memory interfaces, are necessary to efficiently handle memory-intensive tasks like buffering images, streaming high-definition (HD) video, or dealing with compression. Compression also helps with reducing the bandwidth required for wired or wireless data transmission; media processors must rapidly decompress or even decrypt multimedia data, especially if there is any digital rights management (DRM) encryption present."

Smaller the node....

The technology node is at 14nm, even 10nm now. Rei feels that, as we go further down, challenge